# STAT 581: Decision Theory and Bayesian Inference

Handwritten Notes 1

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September 30; October 5, 7, and 12, 2026

**Editorial conventions and sources**

This is an English transcription of `notes/stat581-note1.pdf`, with the original order and substantive questions retained. Source-page ranges appear under section headings. Repeated recaps are consolidated. The final source page (29) is blank. The reference is Yen-Chi Chen, *Lecture 1: Elementary decision theory*, Autumn 2026, `Lec1_decision.pdf`; references below use its printed page numbers. Yellow boxes preserve handwritten TODOs and give proposed resolutions; red boxes identify corrections; blue boxes supply additional explanations. All formulas are displayed in mathematical environments.

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# 1 Elementary decision theory: September 30

## 1.1 The statistical decision problem

*Source: pp. 1–3; lecture pp. 1-1–1-2.*

Nature is an unknown parameter that determines the distribution of observable data. An action is selected after observing those data:

$$\theta \in \Theta, \qquad X \in \mathcal{X}, \quad X \sim P_\theta, \qquad a \in \mathcal{A}. \tag{1}$$

A loss function evaluates the consequence of an action under the true state:

$$L : \Theta \times \mathcal{A} \longrightarrow \mathbb{R}, \qquad (\theta, a) \longmapsto L(\theta, a). \tag{2}$$

In the standard setup, loss has no direct data argument. It becomes random when the action is chosen using random data. Directly minimizing loss over actions is usually infeasible because nature is unknown; a decision rule provides an executable link from observations to actions.

A nonrandomized rule is a measurable map, while a randomized rule is a conditional probability measure (a Markov kernel):

$$\delta : \mathcal{X} \to \mathcal{A}, \qquad D(da \mid x), \qquad A \mid X = x \sim D(\cdot \mid x). \tag{3}$$

The nonrandomized case is the point mass at the selected action:

$$D(B \mid x) = \mathbf{1}\{\delta(x) \in B\}. \tag{4}$$

An MLE, when defined with a measurable choice among maximizers, is an example of a nonrandomized decision rule.

**Supplement: Question in the margin: can loss depend on the data?**

Yes. One can specify a data-dependent loss and define its risk accordingly:

$$r(\theta, D) = \int_{\mathcal{X}} \int_{\mathcal{A}} L(\theta, a, x) D(da \mid x) P_\theta(dx). \tag{5}$$

This is a different decision problem. The lecture uses the simpler loss depending only on nature and the action. Even there, the realized loss depends on the data through the selected action.

## 1.2 Risk averages over the randomness

For nonrandomized and randomized rules, respectively, the risk is

$$r(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))], \tag{6}$$

$$r(\theta, D) = \mathbb{E}_\theta[L(\theta, A)] = \int_{\mathcal{X}} \int_{\mathcal{A}} L(\theta, a) D(da \mid x) P_\theta(dx). \tag{7}$$

The second expression averages over both sampling randomness and conditional action randomness. The loss function is generally specified as part of the problem; the risk additionally depends on the model and the rule.

**TODO: N1-T1: check the randomized-risk integral (source p. 2)**

The handwritten formula treats a probability measure as though it were always a density. The measure form above covers discrete, continuous, and point-mass rules. If both sampling and action densities exist, it becomes

$$r(\theta, D) = \int_{\mathcal{X}} \int_{\mathcal{A}} L(\theta, a) \rho(a \mid x) p_\theta(x) \, da \, dx. \tag{8}$$

For a finite action space, replace the inner integral by a sum. This resolves the TODO's apparent concern about the two layers of randomness. The neighboring phrase "Empirical Theory & Sampled Mean" is retained here as a reading cue; no specific theorem is stated beside it.

The risk is a function of the unknown parameter. A rule minimizing it for every parameter simultaneously need not exist. Two ways to compare rules are to minimize average risk under a prior or to minimize worst-case risk.

## 1.3 Example: bus or walk

*Source: pp. 3–6; lecture Example 1.3, pp. 1-2–1-4.*

Class starts in seven minutes. Walking takes nine minutes; a bus takes three, six, nine, or twelve minutes depending on traffic. The action is zero for the bus and one for walking. The parameter is bus travel time:

$$\mathcal{A} = \{0, 1\}, \qquad \Theta = \{3, 6, 9, 12\}. \tag{9}$$

The observation indicates whether much traffic is visible:

$$X = \begin{cases} 1, & \text{much traffic visible,} \\ 0, & \text{little traffic visible,} \end{cases} \qquad X \sim \text{Bernoulli}(\theta/15). \tag{10}$$

Loss is the number of minutes late. Thus the loss table and the four possible nonrandomized rules are

| $L(\theta, a)$ | $\theta = 3$ | $\theta = 6$ | $\theta = 9$ | $\theta = 12$ |
|---|---|---|---|---|
| $a = 0$ | 0 | 0 | 2 | 5 |
| $a = 1$ | 2 | 2 | 2 | 2 |

(11)

$$\delta_1(x) = 0 \qquad \text{(always take the bus),} \tag{12}$$

$$\delta_2(x) = 1 \qquad \text{(always walk),} \tag{13}$$

$$\delta_3(x) = x \qquad \text{(walk when traffic is visible),} \tag{14}$$

$$\delta_4(x) = 1 - x \qquad \text{(take the bus when traffic is visible).} \tag{15}$$

For any of these rules,

$$r(\theta, \delta) = L(\theta, \delta(1))\frac{\theta}{15} + L(\theta, \delta(0))\left(1 - \frac{\theta}{15}\right). \tag{16}$$

The resulting risks are

| $\theta$ | $r(\theta, \delta_1)$ | $r(\theta, \delta_2)$ | $r(\theta, \delta_3)$ | $r(\theta, \delta_4)$ |
|---|---|---|---|---|
| 3 | 0 | 2 | 0.4 | 1.6 |
| 6 | 0 | 2 | 0.8 | 1.2 |
| 9 | 2 | 2 | 2 | 2 |
| 12 | 5 | 2 | 2.6 | 4.4 |
| maximum | 5 | 2 | 2.6 | 4.4 |

(17)

Always walking is minimax. This remains true when randomization is allowed: at the parameter value nine both actions have loss two, so every rule has worst-case risk at least two, and always walking attains this bound.

Under the uniform prior, the Bayes risks, in rule order, are

$$(r_\pi(\delta_1), r_\pi(\delta_2), r_\pi(\delta_3), r_\pi(\delta_4)) = \left(\frac{7}{4}, 2, \frac{5.8}{4}, \frac{9.2}{4}\right) = (1.75, 2, 1.45, 2.30). \tag{18}$$

The third rule is Bayes under this prior. Randomization cannot lower the minimum: every randomized binary rule is a mixture of the four deterministic rules, and Bayes risk is linear in that mixture.

**Correction: Worst-case risk is not variance (source p. 5)**

The marginal note attributes the poor worst-case behavior of always taking the bus to "variance." The displayed comparison is of maximum expected loss, not variance of loss. Heavy traffic gives that rule loss five, whereas walking always gives loss two. Variance can be a useful intuition about uncertainty, but it is not the minimax criterion here.

# 2 Comparison of rules and testing: October 5

## 2.1 Bayes risk, minimax risk, and admissibility

*Source: pp. 7–9; lecture pp. 1-2 and 1-9–1-10.*

A prior is a probability measure on the parameter space. When it has a density or mass function, that function can be used as a weight. The Bayes risk depends on the rule, with the parameter integrated out:

$$r_\pi(D) = \int_\Theta r(\theta, D)\Pi(d\theta), \qquad D_\pi \in \arg\min_{D\in\mathcal{D}} r_\pi(D). \tag{19}$$

Changing the prior can change the optimal rule. A prior may express uncertainty or belief about nature; its probabilities need not describe repeated sampling of a fixed parameter. A Bayes-optimal rule need not minimize risk separately at each parameter value.

Write maximal risk and the minimax value as

$$r^*(D) = \sup_{\theta\in\Theta} r(\theta, D), \qquad V = \inf_{D\in\mathcal{D}} r^*(D), \qquad D_{\mathrm{MM}} \in \arg\min_{D\in\mathcal{D}} r^*(D). \tag{20}$$

For a class of nonrandomized rules, replace the rule class by that class. The handwritten notation sometimes calls a rule's maximal risk "minimax risk"; the value after optimization is denoted separately here.

A rule is inadmissible if another rule in the same comparison class satisfies

$$r(\theta, D') \le r(\theta, D) \quad \text{for all } \theta \in \Theta, \qquad r(\theta_0, D') < r(\theta_0, D) \quad \text{for some } \theta_0 \in \Theta. \tag{21}$$

Otherwise it is admissible. In the bus example, the third rule dominates the fourth. The first three rules are admissible, even among randomized rules: the first is forced by attaining zero risk at parameter three, the second is forced by attaining risk at most two at parameter twelve, and the third is the unique uniform-prior Bayes rule.

**Correction: A dominated rule can still be Bayes (source pp. 8–9)**

The assertion on p. 8 that the fourth rule can never be Bayes is too strong; the qualification on p. 9 is correct. A prior concentrated at parameter nine makes all

rules Bayes because every action has loss two there. Dominance rules out Bayes optimality when the strict improvement receives positive prior mass (and the risks being compared are finite). In particular, it rules out the fourth rule under the uniform prior.

**Supplement: Margin questions: two meanings of minimax**

The French note on p. 8 says "this is not a minimax lower bound." Here the goal is an exact optimal rule in a specified decision problem. The lecture's Remark 1.2 distinguishes this from minimax rates and lower bounds in nonparametric or high-dimensional statistics. Histogram density estimation is a typical setting for the latter: a rate also requires a distribution class, a loss, and a tuning rule. The margin question alone does not specify such a problem.

## 2.2 Neyman–Pearson hypothesis testing

*Source: pp. 10–11; lecture Section 1.1.1, pp. 1-4–1-5.*

The parameter space is partitioned into null and alternative sets. Action one rejects the null; action zero does not reject it:

$$\Theta = \Theta_0 \cup \Theta_1, \quad \Theta_0 \cap \Theta_1 = \emptyset, \quad \mathcal{A} = \{0, 1\}, \quad \ell_1 > 0. \tag{22}$$

The loss distinguishes the two kinds of error:

$$L(\theta, a) = \mathbf{1}\{a = 1\}\mathbf{1}\{\theta \in \Theta_0\} + \ell_1 \mathbf{1}\{a = 0\}\mathbf{1}\{\theta \in \Theta_1\}. \tag{23}$$

For a randomized test, let its rejection probability conditional on the data be the test function. Its size, type II error, and power are

$$\varphi(x) = \mathbb{P}(A = 1 \mid X = x), \qquad 0 \le \varphi(x) \le 1, \tag{24}$$

$$\text{size}(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi(X)] = \sup_{\theta \in \Theta_0} r(\theta, D), \tag{25}$$

$$\beta(\theta) = \mathbb{E}_\theta[1 - \varphi(X)] = r(\theta, D)/\ell_1, \qquad \theta \in \Theta_1, \tag{26}$$

$$\text{power}_\theta(\varphi) = 1 - \beta(\theta). \tag{27}$$

**TODO: N1-T2: interpret the constrained optimum (source p. 11)**

For a specified alternative, minimize its type II error subject to a uniform bound on type I error:

$$\min_{\varphi} \ \beta(\theta) \quad \text{subject to} \quad \sup_{\vartheta \in \Theta_0} \mathbb{E}_\vartheta[\varphi(X)] \le \alpha, \qquad \theta \in \Theta_1. \tag{28}$$

The two handwritten perspectives are classification and risk mitigation. The constraint controls false positives, while the objective reduces false negatives. For a

composite alternative, one test need not minimize error at every alternative simultaneously. Such a test, when it exists, is uniformly most powerful. The fixed positive error cost does not change the optimizer for a specified alternative.

**The margin note about Fisher and p-values**

The handwritten dates are historical reminders, not assumptions of the decision problem. Fisher's significance-testing perspective emphasizes p-values; the Neyman–Pearson formulation emphasizes error control and power. The claim that p-values occur "only" in Fisher's framework is too restrictive: p-values can also be used to implement level-based tests.

## 2.3 Point estimation and removing randomization

*Source: p. 12 and recap on p. 13; lecture Theorem 1.4, pp. 1-5–1-6.*

In point estimation the action is an estimate of the parameter. For example,

$$X \sim N(\theta, 1), \quad \mathcal{A} = \Theta = \mathbb{R}, \quad \delta(X) = X, \quad L(\theta, a) = (\theta - a)^2. \tag{29}$$

Suppose the action space is convex, loss is convex in the action, and the conditional mean action exists and belongs to the action space. Then the mean action defines a nonrandomized rule with no greater risk:

$$\bar{\delta}(x) = \int_{\mathcal{A}} aD(da \mid x), \qquad r(\theta, \bar{\delta}) \le r(\theta, D) \quad \text{for every } \theta. \tag{30}$$

**TODO: N1-T3: prove the convex-loss result (source p. 12)**

Conditional Jensen's inequality gives

$$L(\theta, \bar{\delta}(x)) = L\left(\theta, \mathbb{E}[A \mid X = x]\right) \tag{31}$$

$$\le \mathbb{E}[L(\theta, A) \mid X = x] = \int_{\mathcal{A}} L(\theta, a) D(da \mid x). \tag{32}$$

Integrate both sides against the sampling distribution to obtain the risk inequality. Under squared loss the difference is exactly

$$r(\theta, D) - r(\theta, \bar{\delta}) = \mathbb{E}_\theta[\mathrm{Var}(A \mid X)] \ge 0. \tag{33}$$

The handwritten averaging formula uses the wrong integration variable: integration is over actions, not observations. "Dominated" here should mean weakly improved unless a strict inequality is also established. The same argument gives the existence of a nonrandomized Bayes rule when a randomized Bayes rule exists under these conditions (lecture Theorem 1.9).

# 3 Bayesian inference and Bayes rules: October 7

## 3.1 Posterior risk and its pointwise minimization

*Source: pp. 13–14; lecture pp. 1-6–1-9, Theorem 1.10.*

For a prior density and a sampling density, the marginal and posterior are

$$m_\pi(x) = \int_\Theta p(x \mid \theta)\pi(\theta)\, d\theta, \qquad \pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{m_\pi(x)}. \tag{34}$$

The posterior formula is used where the marginal density is positive. The distinction between frequentist and Bayesian reasoning is whether the parameter is given a prior distribution, not whether the sampling density uses conditional-bar notation.

For nonnegative loss (or an integrable signed loss), Tonelli's theorem (or Fubini's theorem) gives the Bayes-risk decomposition:

$$\begin{aligned} r_\pi(\delta) &= \int_\Theta \int_{\mathcal{X}} L(\theta, \delta(x)) p(x \mid \theta)\pi(\theta)\, dx\, d\theta && (35)\\ &= \int_{\mathcal{X}} \left\{ \int_\Theta L(\theta, \delta(x))\pi(\theta \mid x)\, d\theta \right\} m_\pi(x)\, dx && (36)\\ &= \int_{\mathcal{X}} Q(\delta(x) \mid x) m_\pi(x)\, dx, && (37)\\ Q(a \mid x) &= \int_\Theta L(\theta, a)\pi(\theta \mid x)\, d\theta. && (38) \end{aligned}$$

If a measurable rule minimizes posterior risk for marginal-almost every observation and has finite Bayes risk, it is Bayes:

$$\delta_\pi(x) \in \arg\min_{a \in \mathcal{A}} Q(a \mid x). \tag{39}$$

Here "pointwise" refers to observations, not to parameter values.

To estimate a real-valued function of the parameter under squared loss, assuming the requisite second moments are finite, use the posterior mean:

$$\begin{aligned} L(\theta, a) &= (f(\theta) - a)^2, && (40)\\ Q(a \mid x) &= \operatorname{Var}(f(\theta) \mid x) + \big(a - \mathbb{E}[f(\theta) \mid x]\big)^2, && (41)\\ \frac{\partial Q(a \mid x)}{\partial a} &= 2(a - \mathbb{E}[f(\theta) \mid x]), && (42)\\ \delta_\pi(x) &= \mathbb{E}[f(\theta) \mid X = x]. && (43) \end{aligned}$$

The variance decomposition proves the result without needing to justify differentiation under the integral.

## 3.2 Beta–binomial conjugacy and shrinkage

*Source: pp. 15–16; lecture Examples 1.5 and 1.12.*

$$\theta \sim \mathrm{Beta}(\alpha, \beta), \quad \alpha, \beta > 0, \qquad X \mid \theta \sim \mathrm{Binomial}(n, \theta). \tag{44}$$

Multiplying likelihood and prior yields

$$\pi(\theta \mid x) \propto \binom{n}{x} \theta^x (1-\theta)^{n-x} \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)} \tag{45}$$

$$\propto \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}, \tag{46}$$

$$\theta \mid X = x \sim \mathrm{Beta}(\alpha + x, \beta + n - x). \tag{47}$$

Conjugacy makes the update interpretable as adding successes and failures to prior parameters. The pseudo-count interpretation depends on convention: the density exponents are the parameters minus one, whereas the posterior-mean weighting uses their sum as prior effective sample size.

Under squared loss, for a positive sample size,

$$\delta_\pi(x) = \frac{x + \alpha}{n + \alpha + \beta} \tag{48}$$

$$= W_n \frac{\alpha}{\alpha + \beta} + (1 - W_n)\frac{x}{n}, \qquad W_n = \frac{\alpha + \beta}{n + \alpha + \beta}. \tag{49}$$

Thus the posterior mean is a convex combination of the prior mean and the MLE. With fixed prior parameters, the difference from the MLE vanishes:

$$\left|\delta_\pi(x) - \frac{x}{n}\right| \le \frac{\alpha + \beta}{n + \alpha + \beta} \longrightarrow 0. \tag{50}$$

**Correction: A shrinkage identity is not the Bernstein–von Mises theorem**

The margin asks whether the same convex-combination identity holds for other distributions. It holds in this beta–binomial model and also in the normal–normal model below; it is not a universal posterior-mean formula. The Bernstein–von Mises reminder on source p. 16 concerns a stronger, suitably scaled normal approximation to the posterior under regularity conditions. The elementary bound above does not prove that theorem, and the normal approximation is not asserted at binomial boundary parameters here.

## 3.3 Normal–normal conjugacy

*Source: p. 17; lecture Example 1.8 and Example 1.17.*

$$\theta \sim N(\mu, \tau^2), \qquad X_i \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \qquad \sigma^2, \tau^2 > 0. \tag{51}$$

For one observation, multiplying the likelihood and prior gives a quadratic exponent:

$$\pi(\theta \mid x) \propto \exp\left[-\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2}\right]. \tag{52}$$

For the full sample, completing the square gives

$$v_n = \frac{1}{\tau^{-2} + n\sigma^{-2}}, \tag{53}$$

$$m_n = \frac{\tau^{-2}\mu + n\sigma^{-2}\bar{x}_n}{\tau^{-2} + n\sigma^{-2}}, \tag{54}$$

$$\theta \mid X_1 = x_1, \dots, X_n = x_n \sim N(m_n, v_n), \qquad \delta_\pi(x) = m_n. \tag{55}$$

Set the sample size to one for the formula in the original black and red writing. The green annotation supplies the full-sample version. The mean is a precision-weighted convex combination of prior and sample means.

**TODO: N1-T4: empirical Bayes (source p. 17)**

The handwritten TODO says "empirical Bayes." In the formulas above the prior hyperparameters are fixed and known. Empirical Bayes estimates some of these hyperparameters from data, often from many related units, and then substitutes them into the posterior rule. The resulting plug-in procedure is not automatically a Bayes rule for the original fixed prior; its risk and uncertainty require a separate analysis. No particular empirical-Bayes model or dataset is specified in the source.

# 4 Minimax rules via Bayes rules

## 4.1 Least favorable priors and the Bayes lower bound

*Source: pp. 18–20; lecture Theorem 1.13 and Corollary 1.14.*

Define the optimal Bayes risk for each prior. A least favorable prior maximizes that optimized value, rather than the risk of an arbitrarily fixed rule:

$$b(\pi) = \inf_{D \in \mathcal{D}} r_\pi(D), \qquad b(\pi_0) = \sup_\pi b(\pi). \tag{56}$$

When a Bayes rule exists, its Bayes risk equals this infimum. For every prior and every rule,

$$\sup_\theta r(\theta, D) \geq r_\pi(D) \geq b(\pi), \qquad \inf_D \sup_\theta r(\theta, D) \geq \sup_\pi b(\pi). \tag{57}$$

Suppose a proper-prior Bayes rule has Bayes risk equal to its finite maximal risk:

$$b(\pi_0) = r_{\pi_0}(D_{\pi_0}) = \sup_\theta r(\theta, D_{\pi_0}) < \infty. \tag{58}$$

Then the rule is minimax and the prior is least favorable.

**TODO: N1-T5a: verify minimaxity (source p. 19)**

For any competitor, the lower-bound chain is

$$\sup_\theta r(\theta, D) \geq r_{\pi_0}(D) \geq r_{\pi_0}(D_{\pi_0}) = \sup_\theta r(\theta, D_{\pi_0}). \tag{59}$$

This proves the required comparison with every rule.

**TODO: N1-T5b: complete least favorability (source p. 19)**

For any other prior, its optimal Bayes risk is no greater than the Bayes risk of the candidate rule under that prior:

$$b(\rho) \leq r_\rho(D_{\pi_0}) \leq \sup_\theta r(\theta, D_{\pi_0}) = b(\pi_0). \tag{60}$$

Taking the supremum over priors proves least favorability. A least favorable prior by itself is not the equality hypothesis in (58); the latter is the sufficient condition used here.

**Constant-risk corollary.** A proper-prior Bayes rule whose risk is the same finite constant at every parameter value is minimax, since its Bayes and maximal risks both equal that constant. Constant risk alone, without a Bayes or other lower-bound argument, is insufficient.

# 5 Minimax examples and admissibility: October 12

## 5.1 A minimax binomial estimator

*Source: pp. 21–22; lecture Example 1.15, pp. 1-10–1-11.*

Return to the beta–binomial model under squared error. Its posterior-mean rule has frequentist risk given by squared bias plus variance:

$$r(\theta, \delta_{\alpha,\beta}) = \left(\theta - \frac{\alpha + n\theta}{n + \alpha + \beta}\right)^2 + \frac{n\theta(1-\theta)}{(n+\alpha+\beta)^2} \tag{61}$$

$$= \frac{\alpha^2 + [n - 2\alpha(\alpha+\beta)]\theta + [(\alpha+\beta)^2 - n]\theta^2}{(n+\alpha+\beta)^2}. \tag{62}$$

Eliminate the linear and quadratic terms by choosing

$$(\alpha+\beta)^2 = n, \qquad 2\alpha(\alpha+\beta) = n, \qquad \alpha = \beta = \frac{\sqrt{n}}{2}. \tag{63}$$

For a positive integer sample size this is a proper beta prior. The rule and its constant risk are

$$\delta_{\mathrm{MM}}(x) = \frac{x + \sqrt{n}/2}{n + \sqrt{n}}, \qquad r(\theta, \delta_{\mathrm{MM}}) = \frac{n/4}{(n+\sqrt{n})^2} = \frac{1}{4(\sqrt{n}+1)^2}. \tag{64}$$

The constant-risk Bayes corollary proves minimaxity. The symmetric beta prior used above is least favorable.

**Correction: Bias–variance formula and scope (source pp. 21–22)**

The variance term is the conditional variance of the estimator, not a variance of its expectation. The question "Can we always do this?" has answer no: there need not be a proper prior yielding a constant-risk Bayes rule. Also, the Bayes rule above is for squared loss; changing loss may change both the Bayes and minimax rules.

## 5.2 Sequences of Bayes risks establish minimaxity

*Source: pp. 23–25; lecture Theorem 1.16 and Example 1.17.*

Suppose proper priors have optimal Bayes risks converging to the finite maximal risk of a candidate rule:

$$b_k = \int_\Theta r(\theta, D_{\pi_k})\Pi_k(d\theta), \qquad \lim_{k\to\infty} b_k = \sup_\theta r(\theta, D_0) < \infty. \tag{65}$$

Then the candidate is minimax.

**TODO: N1-T6: prove the limiting theorem (source p. 24)**

For every competitor and every positive integer index,

$$\sup_\theta r(\theta, D) \geq r_{\pi_k}(D) \geq b_k. \tag{66}$$

Take the limit to obtain

$$\sup_\theta r(\theta, D) \geq \lim_{k\to\infty} b_k = \sup_\theta r(\theta, D_0). \tag{67}$$

Only the numerical Bayes risks must converge; convergence of the rules themselves is not the theorem's assumption. The handwritten limit toward zero is corrected to a limit toward infinity, and the missing prior measure in the Bayes-risk integral is restored.

For independent normal observations with known variance, squared-error risk of the sample mean is constant:

$$X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \quad \theta \in \mathbb{R}, \qquad r(\theta, \bar{X}_n) = \frac{\sigma^2}{n}. \tag{68}$$

Use the proper priors and their normal–normal Bayes risks:

$$\pi_k = N(\mu, k^2), \qquad b_k = \frac{1}{k^{-2} + n\sigma^{-2}} \longrightarrow \frac{\sigma^2}{n}. \tag{69}$$

The sample mean is therefore minimax. For unit sampling variance this is the handwritten sequence with risks tending to the reciprocal sample size.

**Supplement: Why the usual sample mean is not the proper normal-prior rule**

For every finite positive prior variance, the normal–normal posterior mean shrinks toward the prior mean and differs from the sample mean. The sample mean is obtained in a diffuse-prior limit; equivalently it is generalized Bayes for an improper flat prior on the real line. Such a flat prior is not a probability distribution, so the proper-prior constant-risk corollary cannot be applied directly. This answers the Chinese margin question on p. 23 about the normal–normal Bayes rule.

## 5.3 Extending a minimax rule to a larger model

*Source: p. 26; lecture Lemma 1.18 and Example 1.19.*

Suppose a rule is minimax on a smaller model and its maximal risk does not increase on a larger model:

$$\mathcal{P}_1 \subset \mathcal{P}_2, \qquad \sup_{P\in\mathcal{P}_1} r(P, D_0) = \sup_{P\in\mathcal{P}_2} r(P, D_0) = C. \tag{70}$$

For every competitor,

$$\sup_{P\in\mathcal{P}_2} r(P, D) \geq \sup_{P\in\mathcal{P}_1} r(P, D) \geq C. \tag{71}$$

Thus the candidate is minimax on the larger model as well.

For normal mean estimation, take

$$\mathcal{P}_1 = \{N(\theta, \sigma_0^2) : \theta \in \mathbb{R}\}, \tag{72}$$

$$\mathcal{P}_2 = \{N(\theta, \sigma^2) : \theta \in \mathbb{R}, 0 < \sigma^2 \leq \sigma_0^2\}. \tag{73}$$

Both maximal risks of the sample mean equal the known bound divided by sample size.

**Correction: Unknown variance needs a common bound**

The source says only that the unknown variance is finite. The lecture's application requires a fixed finite upper bound over the whole model. Individual finiteness does not supply a finite maximum over an unbounded variance class. With no bound, this argument does not establish a finite minimax value.

## 5.4 Uniqueness and admissibility

*Source: pp. 27–28; lecture Theorems 1.20–1.23, pp. 1-14–1-15.*

A unique finite-risk Bayes rule is admissible when uniqueness is understood up to null sets under every sampling distribution. A unique finite-value minimax rule is also admissible in that sense: any dominating rule would be another minimax rule. Bayes uniqueness merely under the prior marginal needs additional care, since exceptional sets may be possible at some parameter values.

Define the marginal sampling measure under the prior by

$$Q_\pi(B) = \int_\Theta P_\theta(B)\Pi(d\theta). \tag{74}$$

With a convex action space, strictly convex loss in the action (in particular squared loss), a finite Bayes risk, and an existing Bayes rule, strict convexity gives uniqueness marginal-almost surely. To transfer this uniqueness to every sampling distribution, assume

$$Q_\pi(B) = 0 \implies P_\theta(B) = 0 \quad \text{for every measurable } B \text{ and every } \theta, \qquad P_\theta \ll Q_\pi. \tag{75}$$

**TODO: N1-T7: understand the domination assumption (source p. 27)**

The implication in (75) means that a data event impossible under the prior marginal is impossible under every model distribution. If a competitor dominated the Bayes rule, its Bayes risk would be no larger, so Bayes optimality and finiteness make it another Bayes rule. Strict convexity forces the two rules to agree marginal-almost surely; domination then forces agreement under every sampling law, contradicting strict improvement anywhere. This proves admissibility.
In the normal–normal model all relevant sampling and marginal densities are positive everywhere. Therefore the finite-variance posterior mean is unique in this sense and admissible.

**Correction: Posterior mean versus sample mean (source p. 28)**

The normal–normal conclusion is about the proper-prior posterior mean, not the sample mean. Admissibility of the unrestricted one-dimensional normal sample mean requires a separate proof (continued in Notes 2). Admissibility does not follow simply by taking limits of admissible rules.

## 5.5 A half-line parameter restriction changes admissibility

*Source: p. 28; supplementary calculation beyond the lecture's example.*

Consider the one-observation problem under squared error with real-valued actions:

$$X \sim N(\theta, 1), \quad \theta \geq 0, \qquad \delta(X) = X, \quad \delta_+(X) = \max(0, X). \tag{76}$$

Projection improves squared error whenever the observation is negative:

$$(X-\theta)^2 - (\delta_+(X)-\theta)^2 = (X^2 - 2\theta X)\mathbf{1}\{X < 0\} > 0 \quad \text{on } \{X < 0\}. \tag{77}$$

For every finite nonnegative parameter that event has positive probability. Consequently,

$$r(\theta, \delta_+) < r(\theta, \delta) = 1, \qquad \lim_{\theta\to\infty} r(\theta, \delta_+) = 1, \qquad \sup_{\theta\geq 0} r(\theta, \delta_+) = 1. \tag{78}$$

Both rules are minimax in the real-action problem, but the unprojected rule is inadmissible. One lower-bound justification for the minimax value is to put a uniform prior on a long interval in the half-line. As the interval length grows, the posterior is arbitrarily close to the unit-variance normal posterior away from the interval's endpoints, so the optimal Bayes risks tend to one. These risks bound the minimax value from below, while the constant-risk rule bounds it from above.

**Supplement: A quantitative version of the half-line lower bound**

Use a uniform prior on the interval from zero to a positive length. The posterior is a unit-variance normal centered at the observation, truncated to that interval. If the observation is at least a fixed distance from both endpoints, its posterior variance is at least one minus a quantity that tends to zero as that distance grows. This follows uniformly from the vanishing first and second moments of the two normal tails. Writing the interval length as a capital letter, one obtains

$$b_L = \mathbb{E}[\operatorname{Var}(\theta \mid X)] \geq (1 - \eta_K)\left(1 - \frac{4K}{L}\right)\mathbb{P}(|Z| \leq K), \tag{79}$$

$$L > 4K, \quad Z \sim N(0,1), \quad \eta_K \longrightarrow 0. \tag{80}$$

Indeed, a parameter at least twice that distance from the endpoints and noise no larger than that distance ensure an interior observation. First let the interval length tend to infinity, then let the distance tend to infinity. The lower bound tends to one. The Bayes risk cannot exceed one because the unprojected rule has risk one, so these proper priors establish the claimed minimax lower bound.

**Correction: Equal maximal risk does not mean equal pointwise risk**

The handwritten equality of the two risk functions is false; their suprema agree but the projected risk is strictly smaller at each finite parameter. In the analogous sample-mean problem the common supremum is sampling variance divided by sample size. If actions are restricted to the nonnegative half-line as well, the unprojected rule is not even an allowed rule. The comparison above explicitly uses real-valued actions.

**Continuation**

For the unrestricted real parameter space, the one-dimensional normal sample mean is admissible under squared error (lecture Theorem 1.23). The handwritten proof and the higher-dimensional James–Stein comparison are transcribed in `stat581-note2.tex`.